

IMAGE COVOLUTION

Convolution

- ▶ Convolution is the process in which each element of the image is added to its local neighbors, and then it is weighted by the kernel. It is related to a form of mathematical convolution.
- ▶ In Convolution, the matrix does not perform traditional matrix multiplication but it is denoted by $*$.
- ▶ Convolution is used for many things like calculating derivatives, detect edges, apply blurs etc. and all this is done using a "**convolution kernel**". A convolution kernel is a very small matrix and in this matrix, each cell has a number and also an anchor point.
- ▶ The anchor point is used to know the position of the kernel with respect to the image. It starts at the top left corner of the image and moves on each pixel sequentially. Kernel overlaps few pixels at each position on the image. Each pixel which is overlapped is multiplied and then added. And the sum is set as the value of the current position.

What is mask?

- Mask is also a signal. It can be represented by a two dimensional matrix. The mask is usually of the order of 1×1 , 3×3 , 5×5 , 7×7 . A mask should always be in odd number, because otherwise you cannot find the mid of the mask. Why do we need to find the mid of the mask

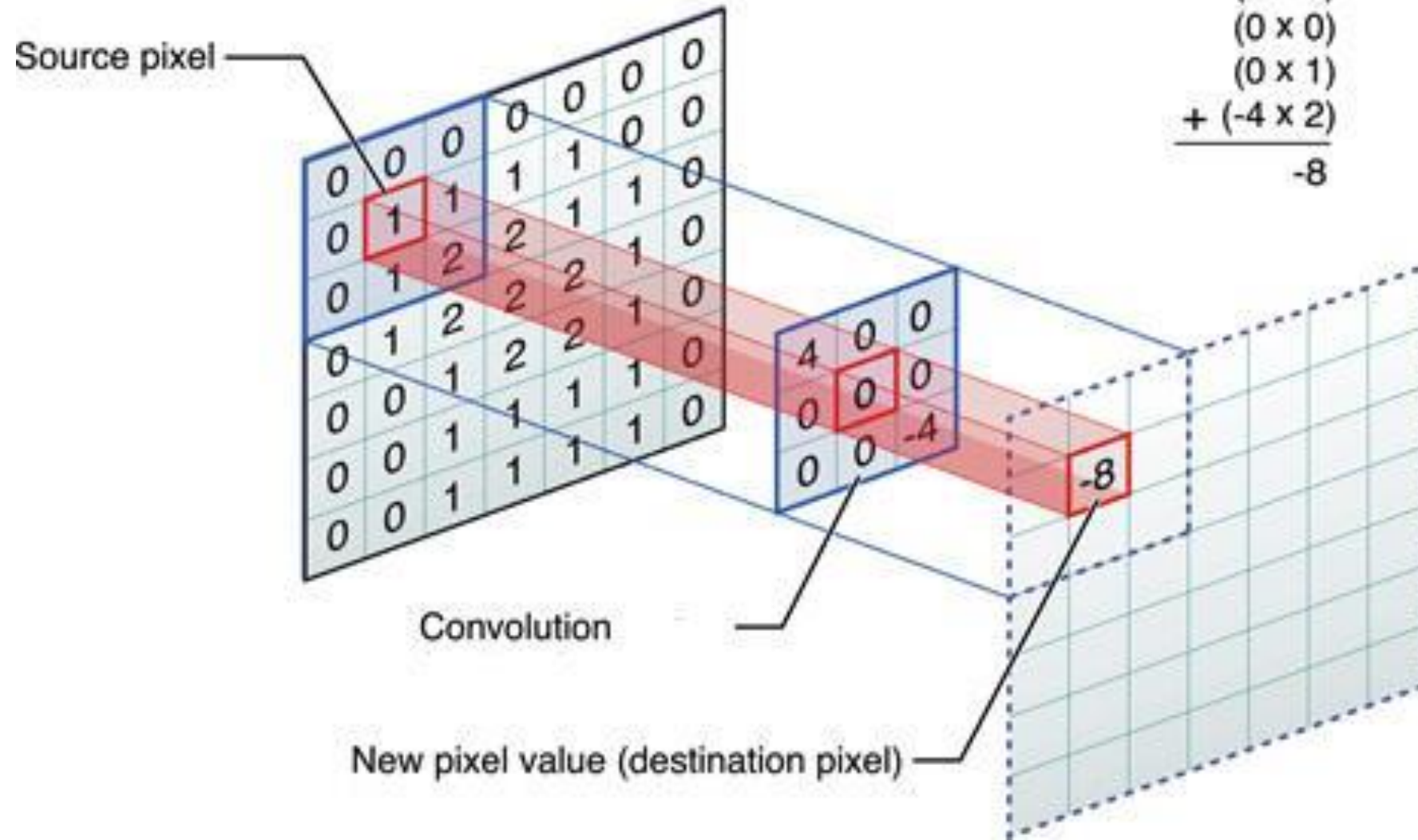
How to perform convolution?

In order to perform convolution on an image, following steps should be taken.

- ▶ Flip the mask (horizontally and vertically) only once
- ▶ Slide the mask onto the image.
- ▶ Multiply the corresponding elements and then add them
- ▶ Repeat this procedure until all values of the image has been calculated.

Center element of the kernel is placed over the source pixel. The source pixel is then replaced with a weighted sum of itself and nearby pixels.

$$\begin{array}{r}
 (4 \times 0) \\
 (0 \times 0) \\
 (0 \times 0) \\
 (0 \times 0) \\
 (0 \times 1) \\
 (0 \times 1) \\
 (0 \times 0) \\
 (0 \times 1) \\
 \hline
 + (-4 \times 2) \\
 \hline
 -8
 \end{array}$$



Why Convolution?

- Convolution can achieve something, that the previous two methods of manipulating images can't achieve. Those include the blurring, sharpening, edge detection, noise reduction e.t.c.

Mathematical Form of Convolution

$$f * h = \sum_k \sum_l f(k, l) h(-k, -l)$$

f = Image

h = Kernel

Mathematical Form of Convolution

$$f * h = \sum_k \sum_l f(k,l)h(-k,-l)$$

f = Image
 h = Kernel

h

h_7	h_8	h_9
h_4	h_5	h_6
h_1	h_2	h_3

$X - flip$

h_1	h_2	h_3
h_4	h_5	h_6
h_7	h_8	h_9

f

f_1	f_2	f_3
f_4	f_5	f_6
f_7	f_8	f_9

$Y - flip$

h_9	h_8	h_7
h_6	h_5	h_4
h_3	h_2	h_1

$*$

$$f * h = f_1h_9 + f_2h_8 + f_3h_7 + f_4h_6 + f_5h_5 + f_6h_4 + f_7h_3 + f_8h_2 + f_9h_1$$

Example

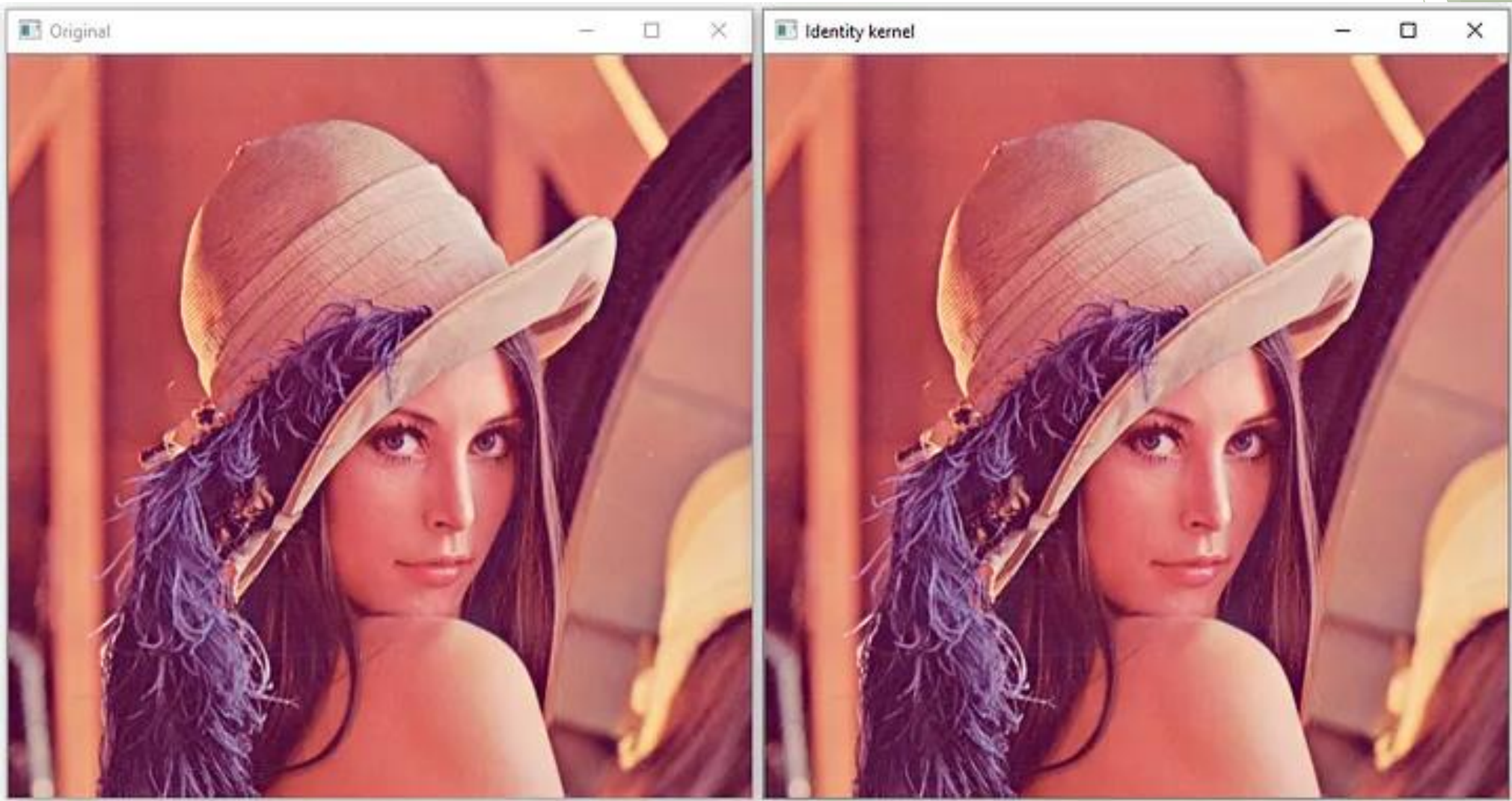
- ▶ In mathematics, convolution is an operation performed on two functions (f and g) to produce a third function. Convolution is one of the most important operations in signal and [image processing](#). It could operate in 1D (e.g. speech processing), 2D (e.g. [image processing](#)) or 3D (video processing).
- ▶ In image processing, convolution is the process of transforming an image by applying a kernel over each pixel and its local neighbors across the entire image. The kernel is a matrix of values whose size and values determine the transformation effect of the convolution process.
- ▶ The Convolution Process involves these steps. (1) It places the Kernel Matrix over each pixel of the image (ensuring that the full Kernel is within the image), multiplies each value of the Kernel with the corresponding pixel it is over. (2) Then, sums the resulting multiplied values and returns the resulting value as the new value of the center pixel. (3) This process is repeated across the entire image.

Continue

- ▶ As we see in the picture, a 3x3 kernel is convoluted over a 7x7 source image.
- ▶ Center Element of the kernel is placed over the source pixel.
- ▶ The source pixel is then replaced with a weighted sum of itself and surrounding pixels.
- ▶ The output is placed in the destination pixel value.
- ▶ In this example, at the first position, we have 0 in source pixel and 4 in the kernel. 4×0 is 0, then moving to the next pixel we have 0 and 0 in both places. 0×0 is 0. Then again 0×0 is 0. Next at the center there is 1 in the source image and 0 in the corresponding position of kernel. 0×1 is 0. Then again 0×1 is 0. Then 0×0 is 0 and 0×1 is 0 and at the last position it is -4×2 which is -8. Now summing up all these results we get -8 as the answer so the output of this convolution operation is -8. This result is updated in the Destination image.

Continue

- ▶ The output of the convolution process changes with the changing kernel values.
- ▶ For example, an Identity Kernel shown below, when applied to an image through convolution, will have no effect on the resulting image. Every pixel will retain its original value as shown in the following figure.



Continue

- ▶ A Sharpen Kernel like this when applied to an image through convolution, will have an image sharpening effect to the resulting image. The precise values can be customized for varying levels of sharpness as shown in the following figure.

```
[ [-1  -1  -1 ]  
  [-1   9  -1 ]  
  [-1  -1  -1 ]]
```